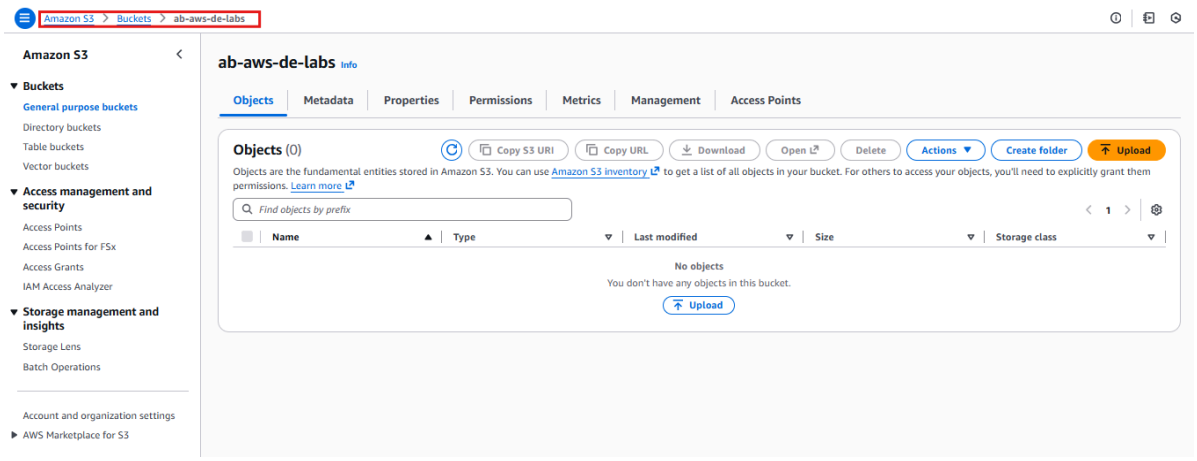
Lab: Orchestrate AWS Glue Jobs Using Apache Airflow

Objective

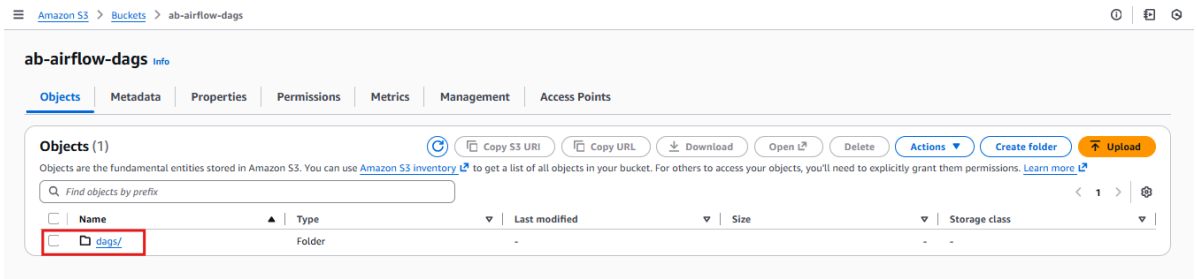
Orchestrate AWS Glue Spark and Python Shell jobs using an Apache Airflow DAG with conditional execution, job monitoring, file archival, and DynamoDB validation.

Step 1: Deploy the Airflow DAG

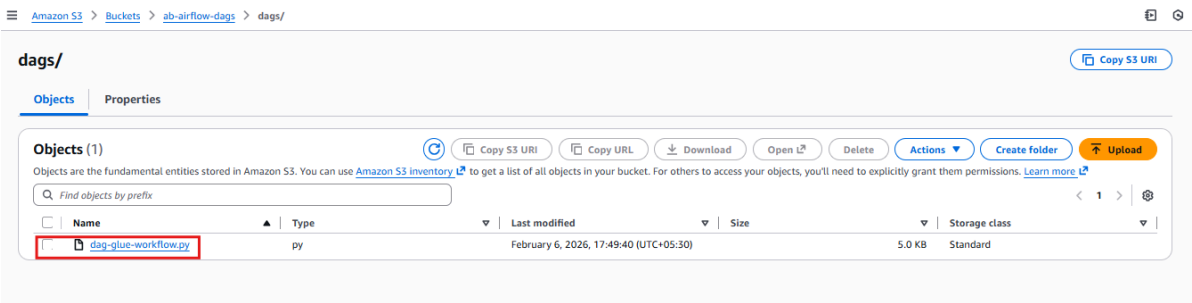
1. Navigate to the S3 bucket used by the Airflow environment.



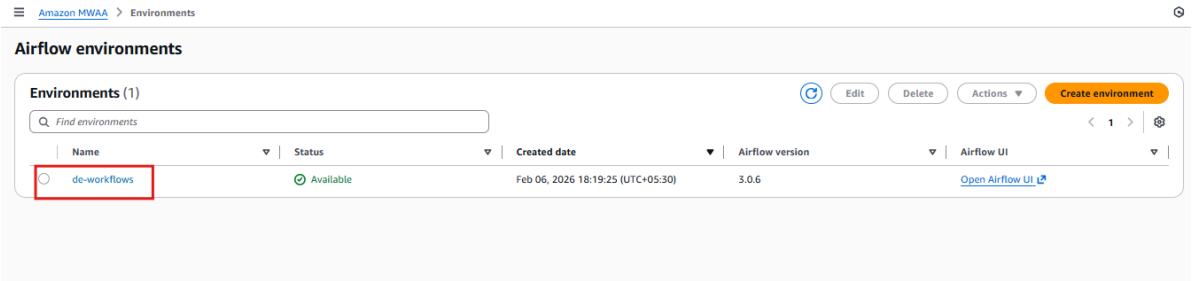
2. Open the **dags** directory.



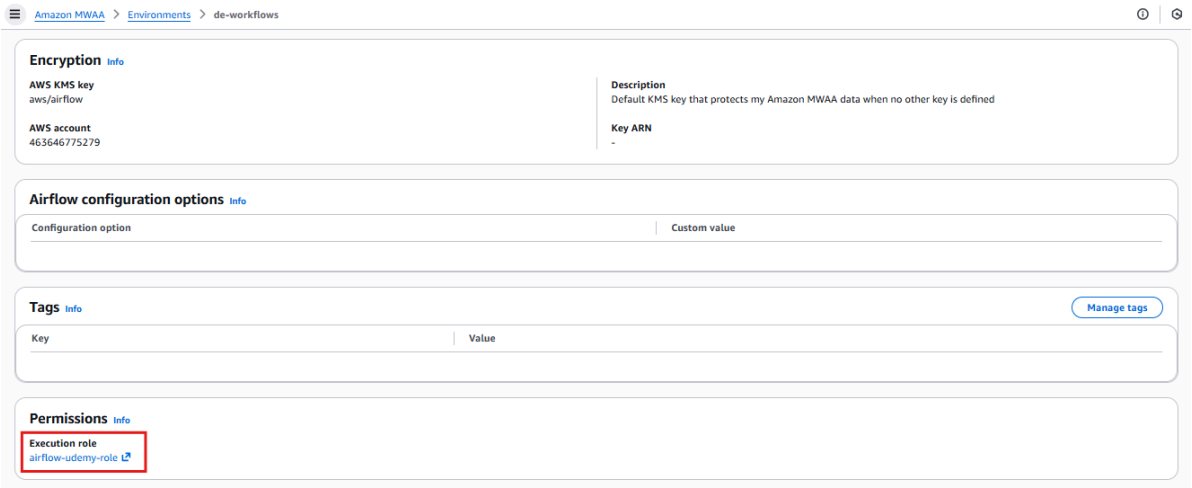
3. Upload the file `dag_glue_workflow.py`.



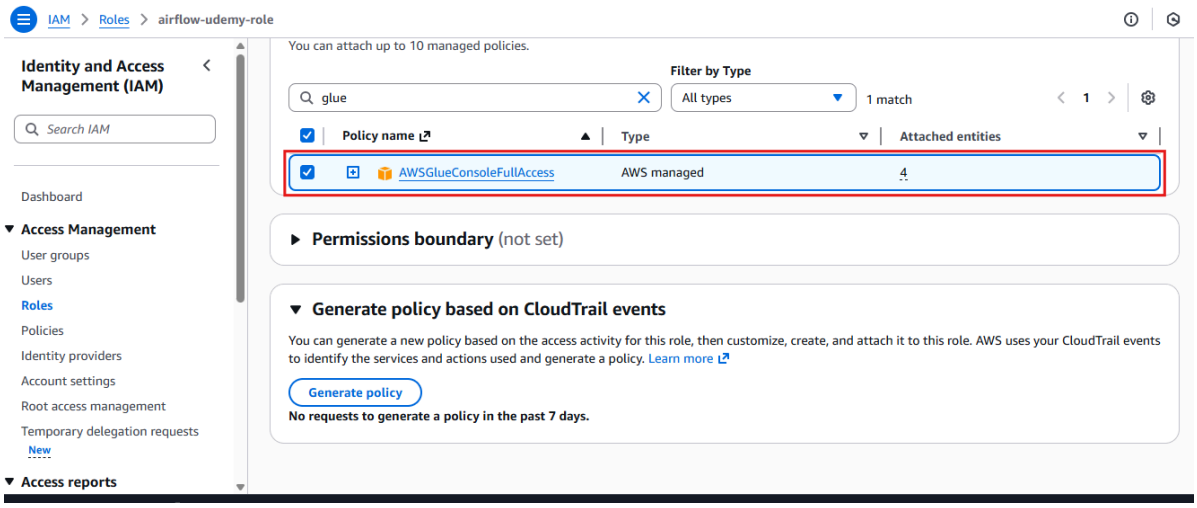
4. Navigate to the Airflow environment in the AWS Console.



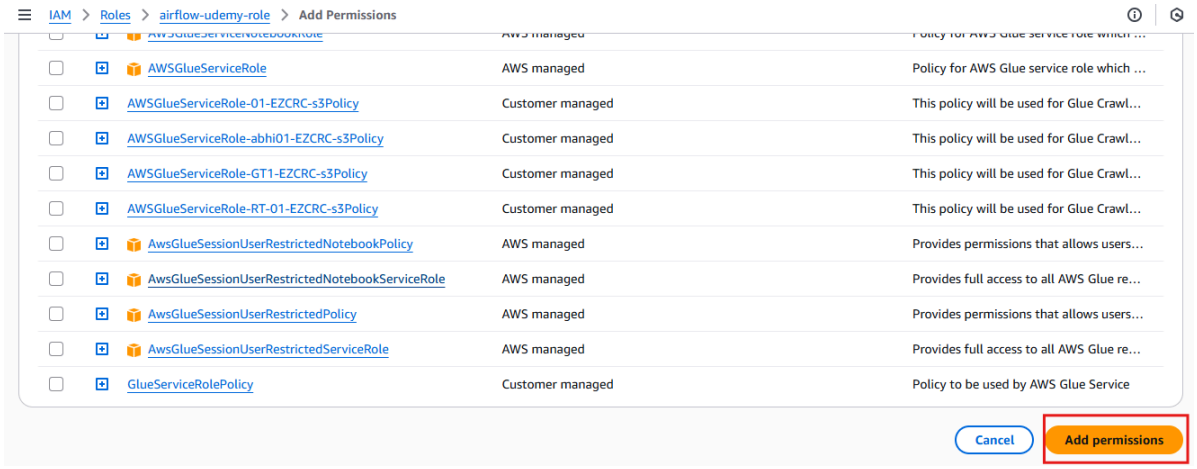
5. Open the execution IAM role associated with Airflow.



6. Attach the **AWSGlueConsoleFullAccess** policy to the role.

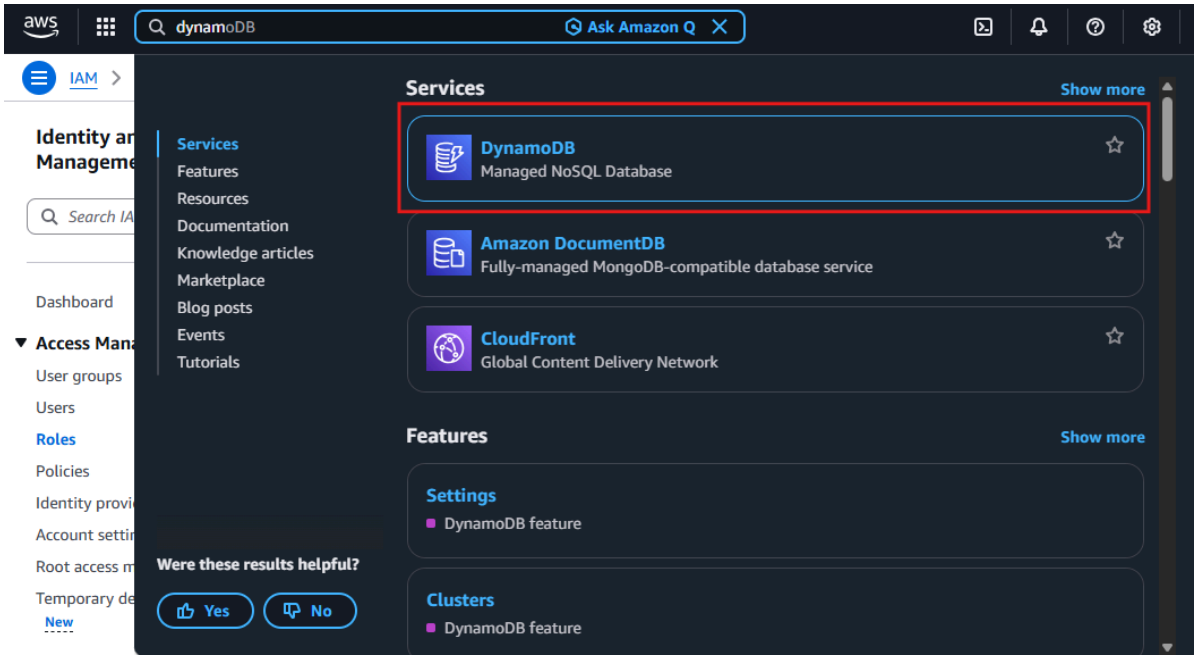


7. Save the updated permissions.

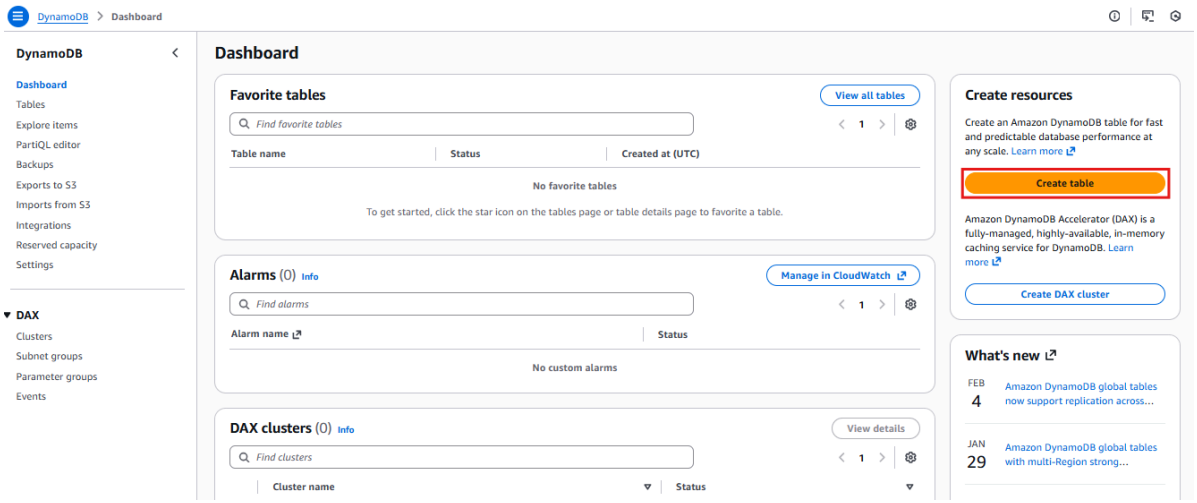


Step 2: Create DynamoDB Table

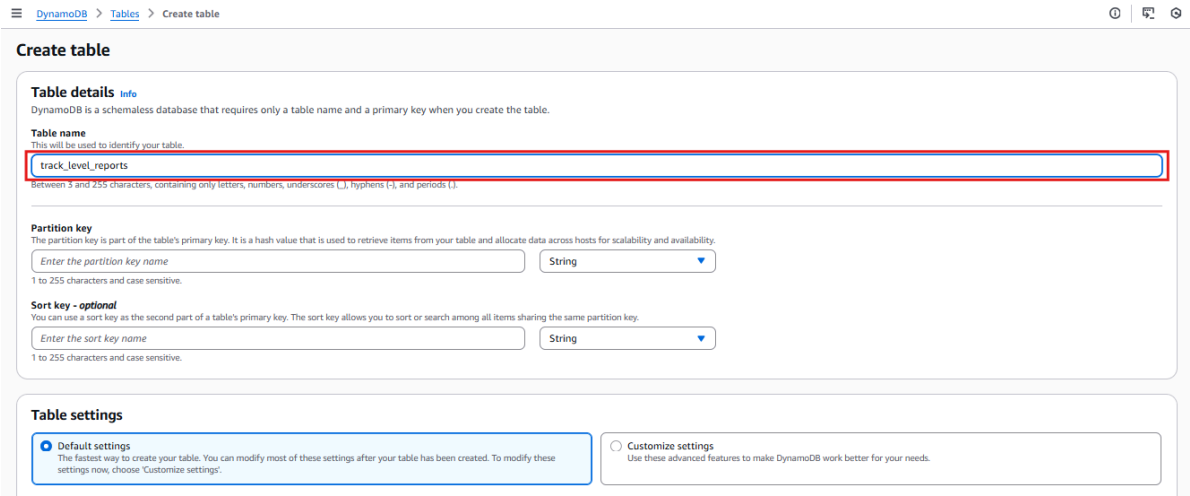
1. Navigate to the **DynamoDB** service.



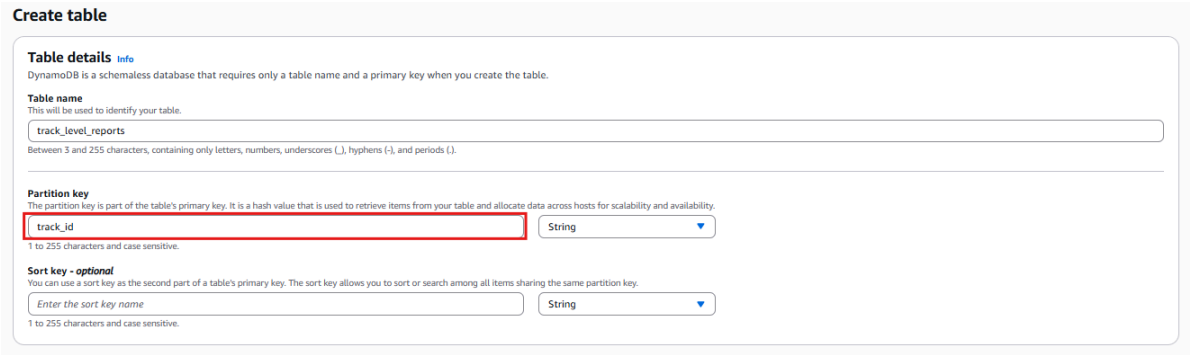
2. Click **Create table**.



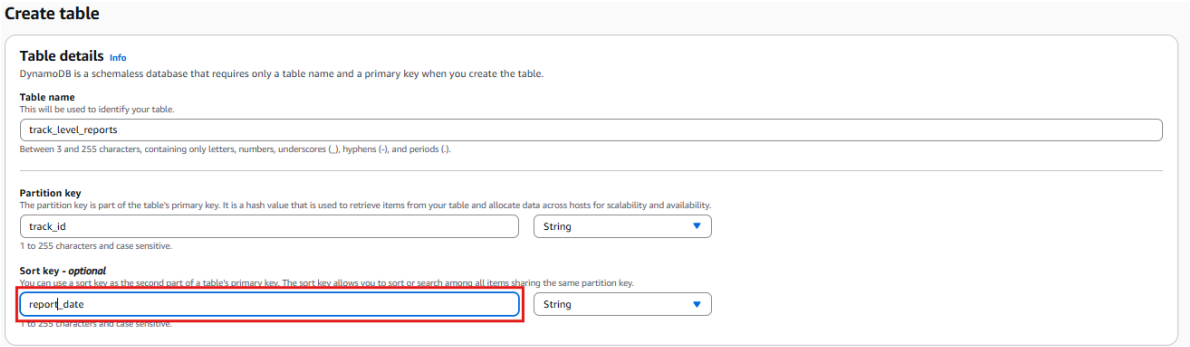
3. Set the table name to **TrackLevelReports**.



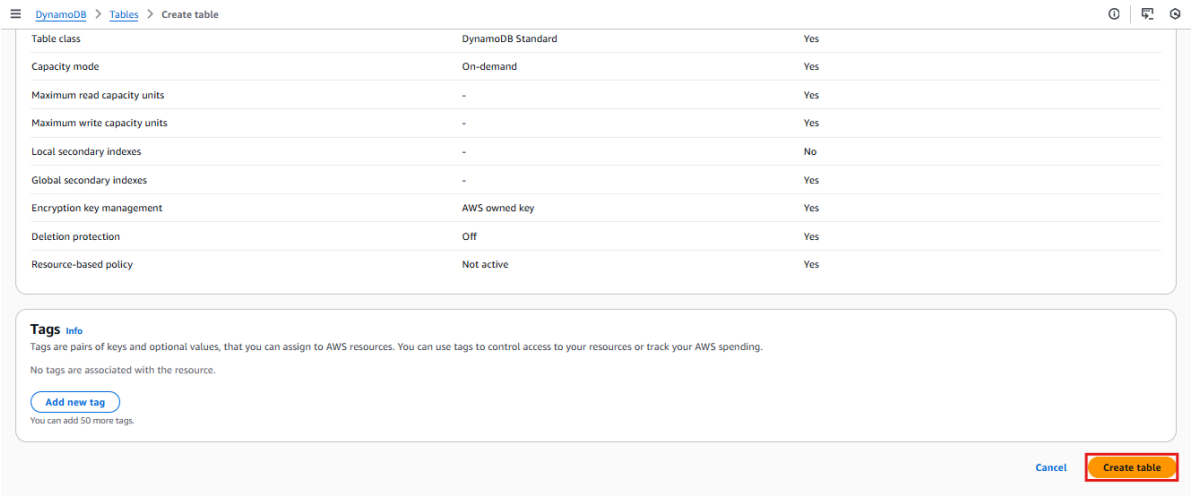
4. Configure **track_id** as the partition key.



5. Configure **report_date** as the sort key.

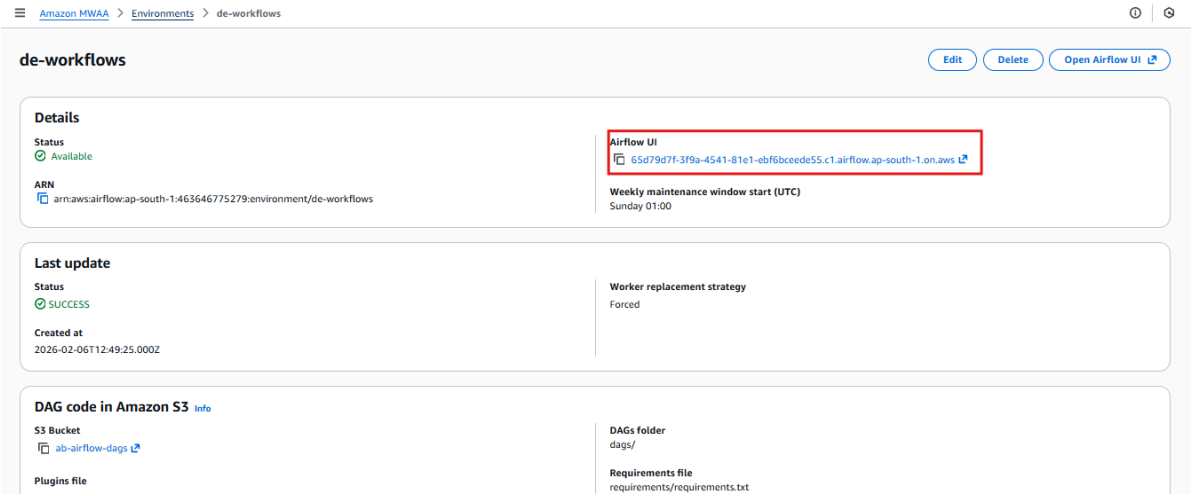


6. Leave all other settings as default.
7. Create the table.

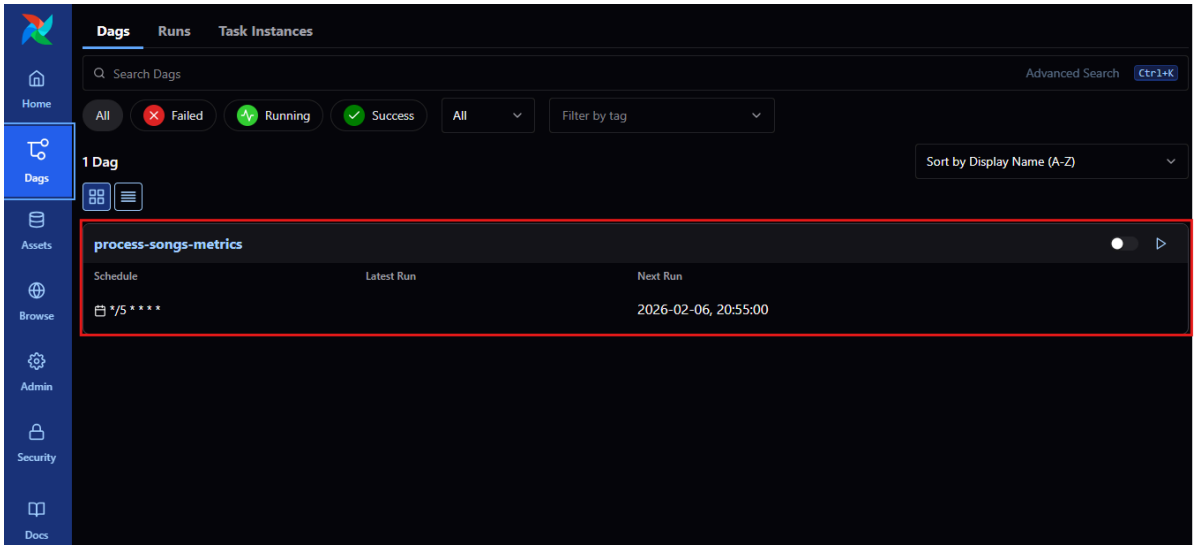


Step 3: Execute and Validate the Workflow

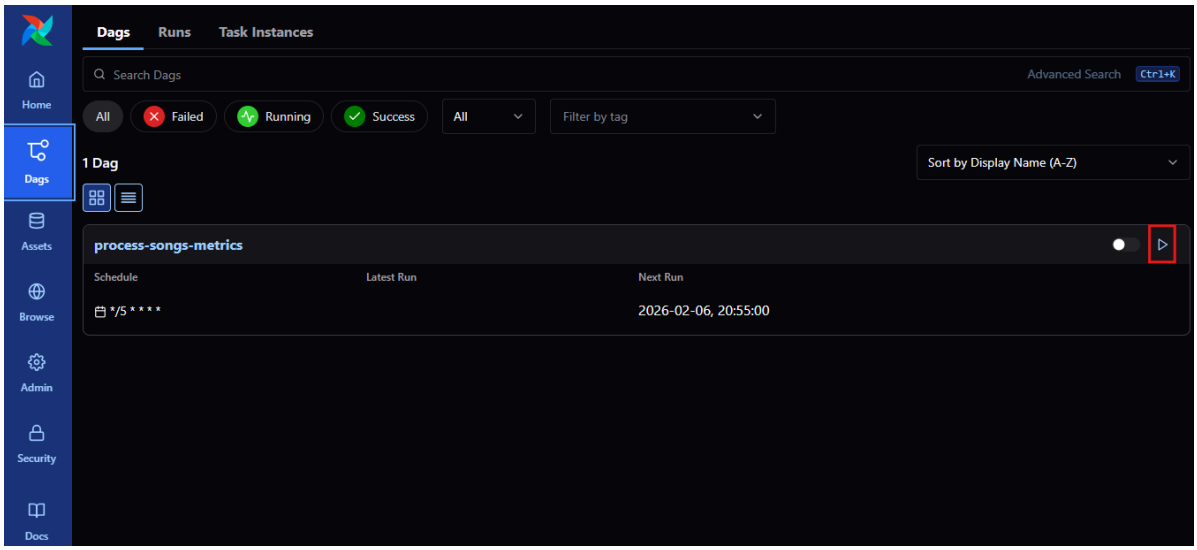
1. Open the Airflow UI.



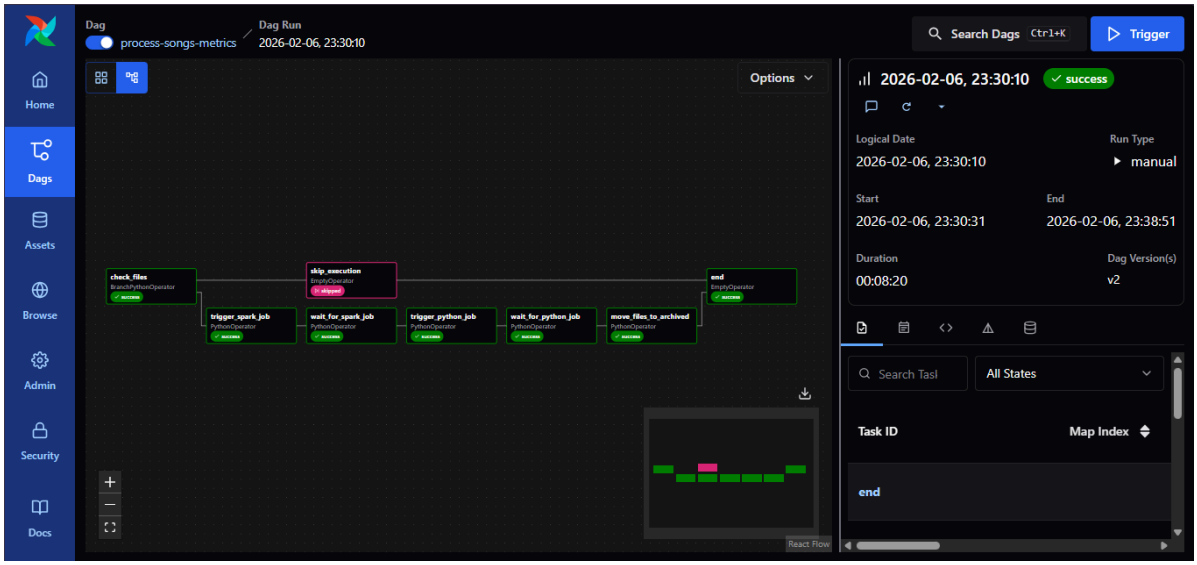
2. Enable the DAG.



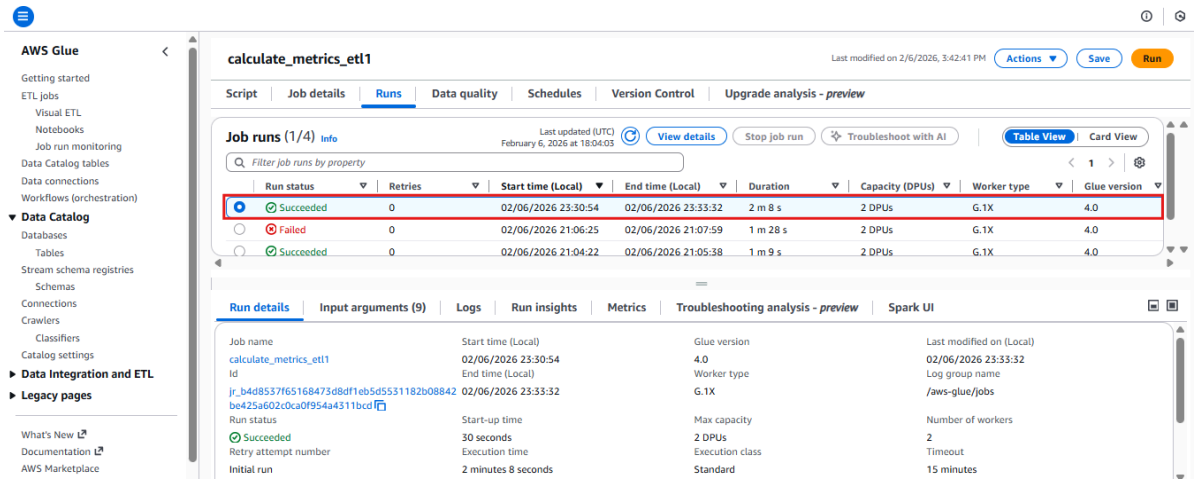
3. Trigger the DAG manually.



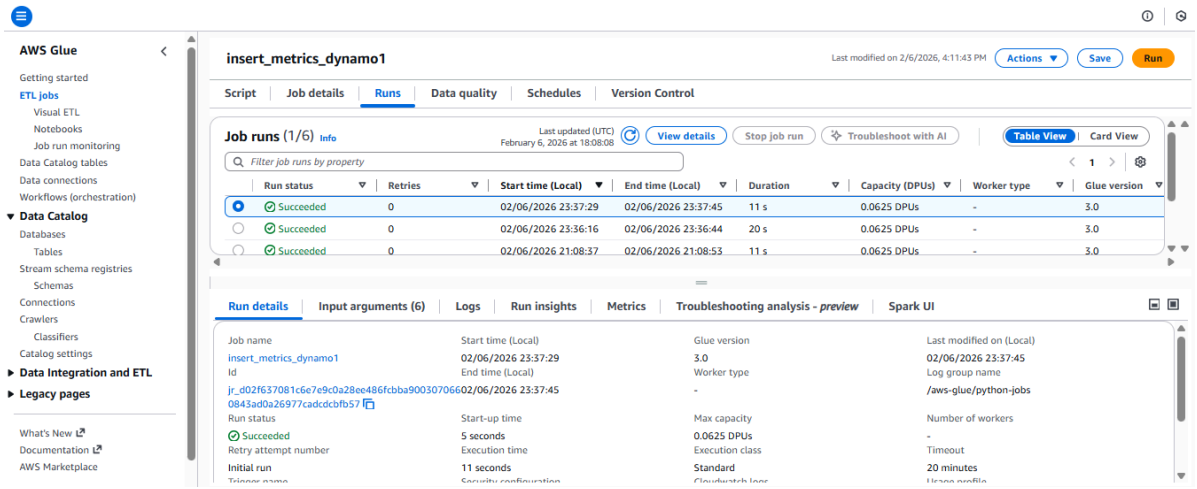
4. Monitor task execution using the **Graph** view.



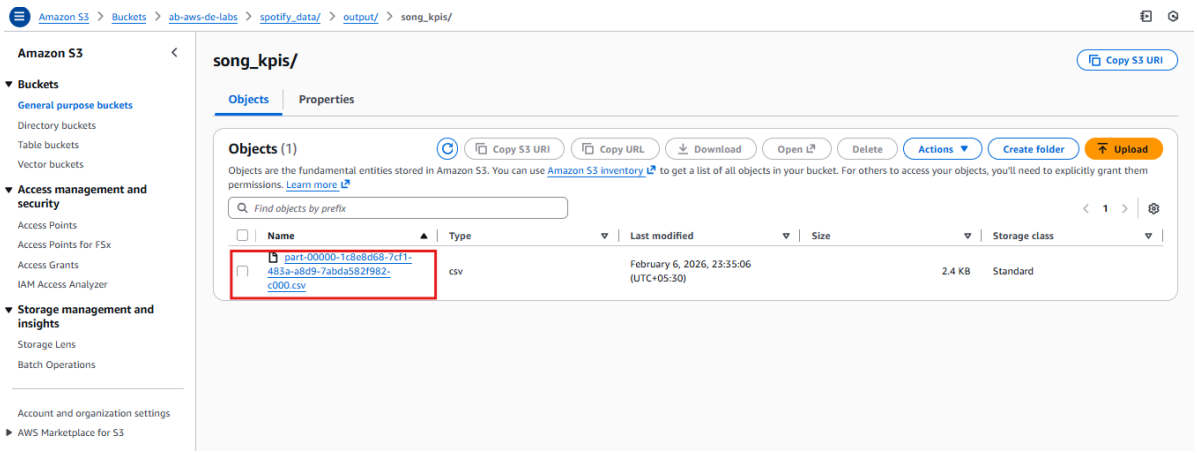
5. Verify Spark Glue job execution from the Glue console.



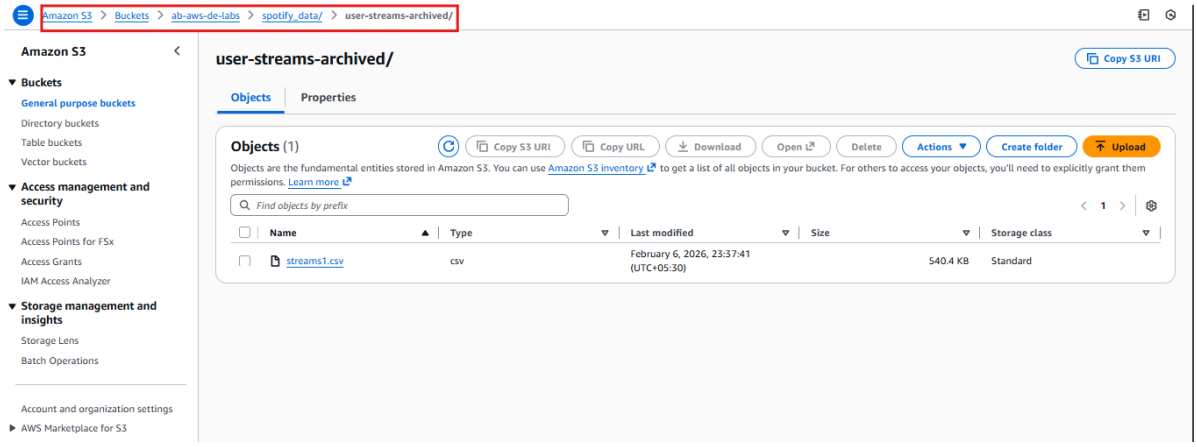
6. Verify Python Shell Glue job execution after Spark completion.



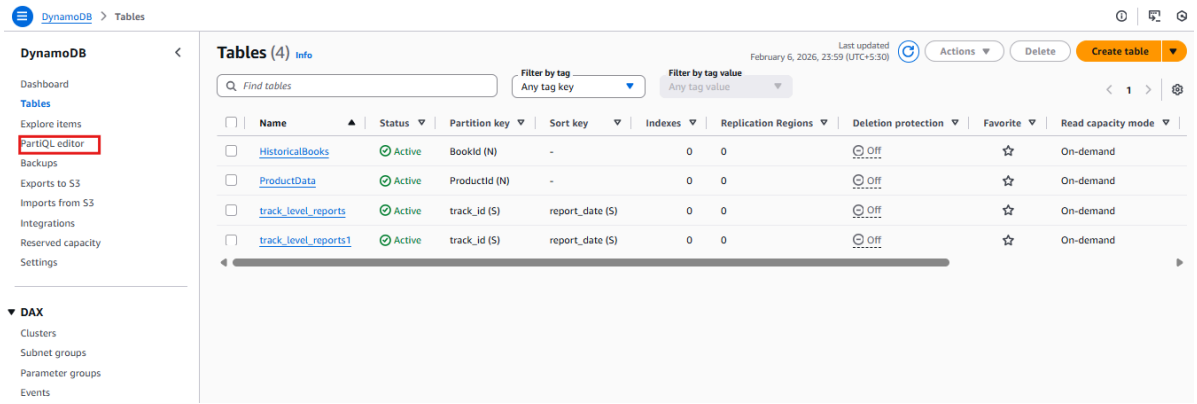
7. Confirm output files are written to the S3 output directory.

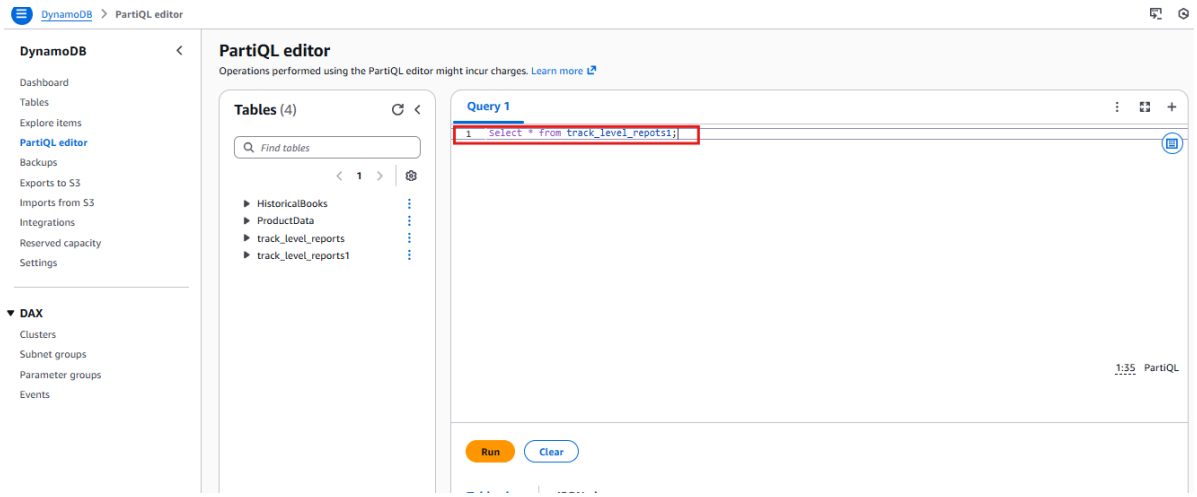


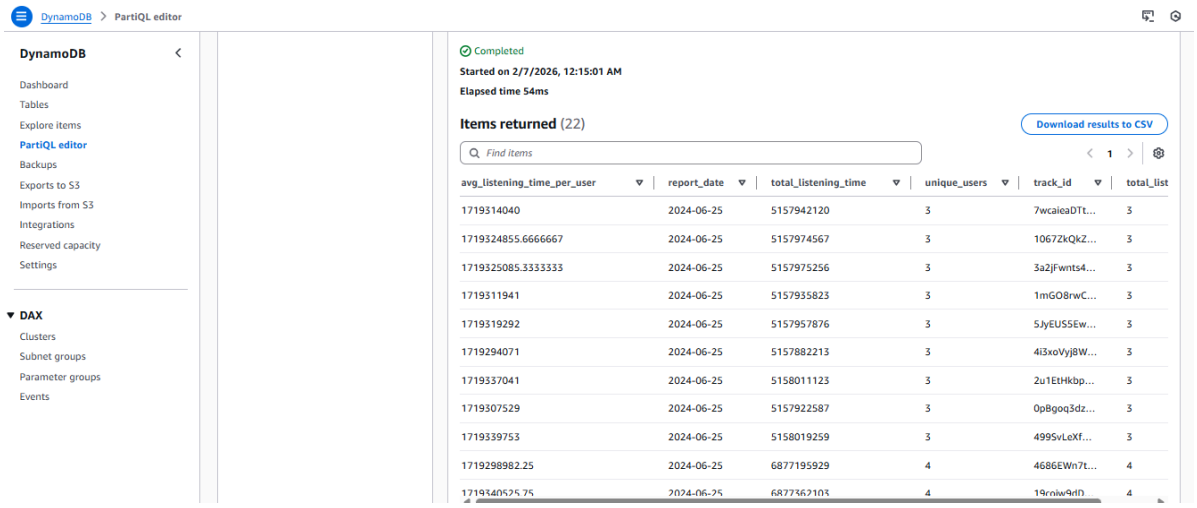
8. Verify user stream files are moved to the archive folder.



9. Query the DynamoDB table to validate inserted records.







Outcome

The Airflow DAG successfully orchestrates AWS Glue Spark and Python Shell jobs. Input files are conditionally processed, results are written to S3 and

DynamoDB, and processed files are archived automatically.